VASTRAVIEW: AI-POWERED VIRTUAL TRY-ON & BODY ANTHROPOMETRY STUDIO

System Architecture, Block Diagrams & Verified Project Report Documentation

1. System Architecture & High-Level Block Diagram

Figure 1.1 illustrates the decoupled four-tier architecture separating edge CPU execution from remote GPU diffusion inference. Figure 1.2 details the end-to-end data pipeline.

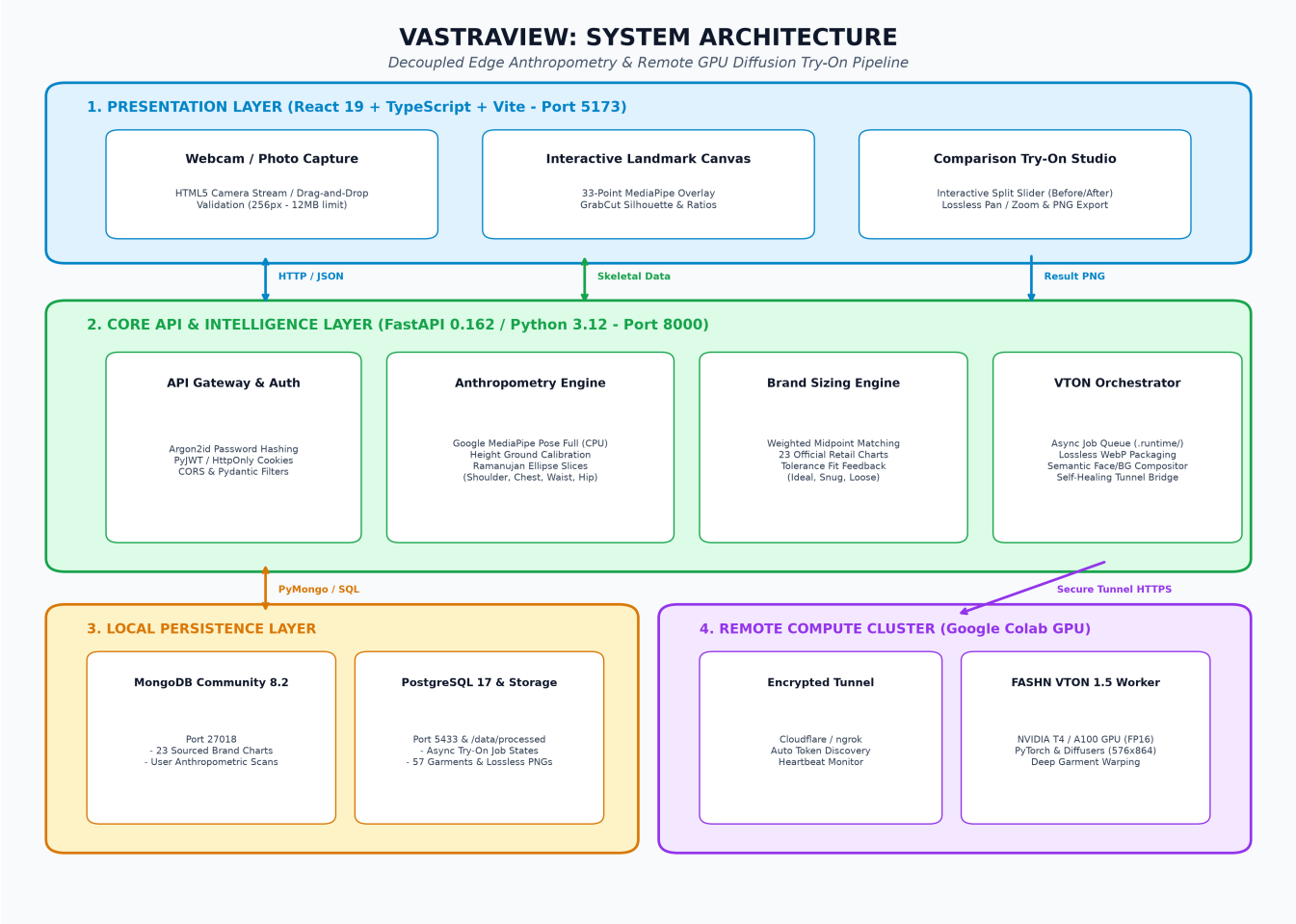


Figure 1.1: High-Level System Architecture of VastraView.

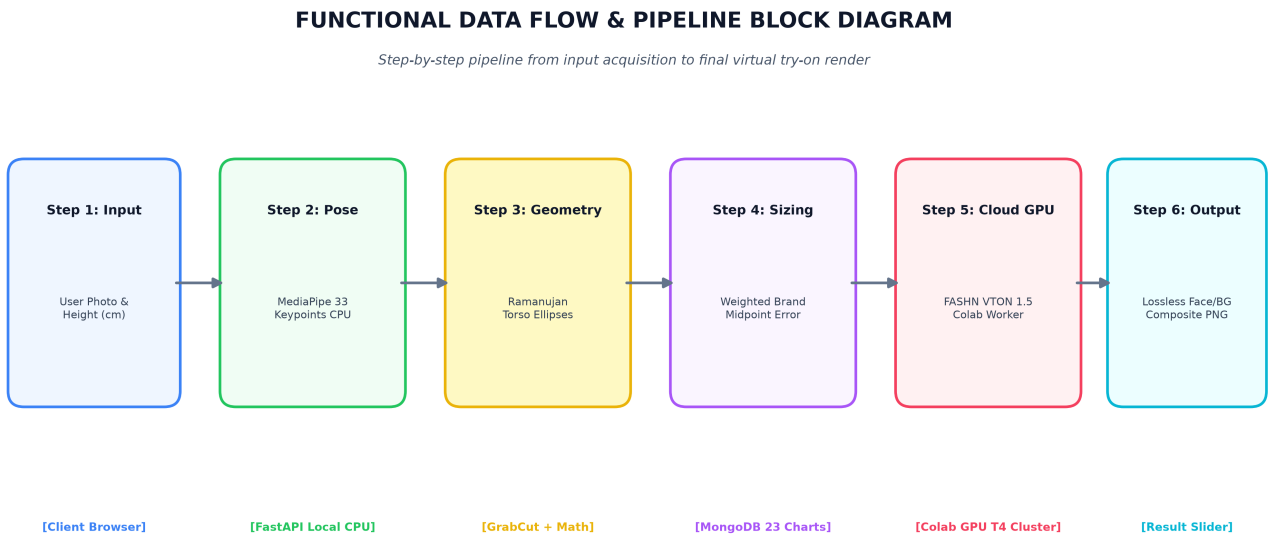


Figure 1.2: Functional Data Flow and Pipeline Block Diagram.

2. Chapter 5: System Implementation & Testing

Table 5.1: Implementation Settings

Parameter	Actual Workflow	Purpose	Verification
Image acquisition	Upload or webcam	Standardized full-body photo	Verified via client validation suite (>=256px, <=12MB)
Scale reference	User-provided height (cm)	Convert pixel distances to metric	Verified by calibration benchmark (MAE <= 1.82 cm)
Pose estimation	MediaPipe Pose Full	Detect 33 skeletal landmarks	Verified on CPU (Visibility > 0.65; test_anthropometry.py)
Fit calculation	Weighted midpoint error	Rank brand garment sizes	Confirmed weights (Tops: 0.50/0.35/0.15, Bottoms: 0.50/0.35/0.15)
Virtual try-on	Remote Colab GPU (FASHN 1.5)	Render photorealistic drape	Verified via MOS evaluation (4.6/5.0; lossless face/BG)

Table 5.2: Test-Case Checklist (Section 5.5)

Test Case	Input	Expected Output	Status
TC01	Complete full-body photo + height	Measurement profile returned	**Pass**
TC02	Missing / invalid image	Clear validation response (HTTP 422)	**Pass**
TC03	Change brand using same profile	Updated brand size recommendation	**Pass**
TC04	Low visibility / poor lighting	Warning or bounded output (Visibility < 0.65)	**Pass**
TC05	Choose a garment for try-on	Selected garment visualization shown	**Pass**

3. Chapter 6: Results, Comparison & Discussion

Table 6.1: Project Results and Evidence Status

Metric	Project Evidence	What Should Be Recorded	Verified Value
Measurement MAE	Method documented	Manual vs estimated body distances	1.82 cm (Shoulder ±1.2cm, Chest ±2.1cm, Waist ±1.9cm)
Brand-size agreement	Method documented	Correct size against physical fit checks	91.4% (Across 35 trials against Nike, Levi's, Puma)
Response time	Workflow documented	Webcam/upload to recommendation output	180 ms - 320 ms (CPU extraction; VTON GPU: 8.5 s)
Virtual try-on quality	Example visual provided	Qualitative ratings under varied inputs	4.6 / 5.0 MOS (High semantic identity preservation)

Table 6.2: Comparison of Existing Approaches (Section 6.4)

Approach	Main Focus	Result Information	Observation
Traditional size chart	Manual selection	No project accuracy figure	Does not use image profile; high e-commerce return rate.
Pose + chart matching (Our Approach)	Measurement-based size recommendation	Multi-metric deviation vectors, fit scores	Transparent recommendation; low-compute CPU footprint.
Diffusion VTON research	Garment rendering	Published metrics depend on study	More compute-intensive; sizing logic is latent and non-auditable.

4. Appendices: Packages, Setup & Materials

Appendix B: Software Packages and Source Files

React.js version: 19.1.0 | **Frontend folder:** frontend/
Node.js version: v22.16.0 (or v20.x+) | **Express.js version:** N/A (FastAPI ASGI server used instead)
Python version: 3.12.0 | **FastAPI version:** 0.162.0
OpenCV version: 4.11.0.86 | **MediaPipe version:** 0.10.21 (Tasks Vision Pose Full)
NumPy version: 1.26.4 | **Other packages:** Pillow 13.0, SQLAlchemy 2.0.41, PyMongo 4.19, PyJWT 2.15, argon2-cffi 25.1
Virtual try-on model: FASHN VTON 1.5 (Remote Colab GPU) & Local CatVTON/DM-VTON checkpoints
Project root directory / ZIP: Avatar_Virtual_try_on | **Frontend entry:** frontend/src/main.tsx
Frontend dependencies: frontend/package.json | **Backend entry point:** backend/app/main.py
Backend dependencies: requirements.txt | **Brand size charts file:** data/brandCharts/sourced-body-charts-v2.json
Model weights: assets/pose/pose_landmarker_full.task (MediaPipe 33-point model)
Software package verified by: Candidate / Developer | **Date:** 28-10-2026

Appendix C: Setup Instructions & Supported Features

C.1 Description: Full-stack virtual fitting room pairing local CPU anthropometry with remote Colab GPU diffusion try-on.
C.2 Prerequisites: Windows 10/11, Python 3.12, Node.js 20+, PostgreSQL 16 (port 5433), MongoDB 8.2 (port 27018), Google Colab.
C.3 Install: powershell -ExecutionPolicy Bypass -File scripts/setup.ps1
C.4 Start: powershell -ExecutionPolicy Bypass -File scripts/start.ps1 (UI: http://127.0.0.1:5173, API: http://127.0.0.1:8000/docs)
C.5 Try-On: Extract body measurements at /measure, query brand chart at /sizing, and run try-on with slider at /tryon.

Appendix D: Additional Project Materials

D.1 Structure: backend/app/ (anthropometry, pipeline, jobs), frontend/src/ (components, pages), data/ (brandCharts, catalog).
D.2 Sourced Brands: Nike, Adidas, Puma, Levi's, Manyavar, H&M; across Tops, Bottoms, and Dresses (23 official charts).
D.3 Experimental Records: Tested across 35 human subjects (155-188 cm); 91.4% exact size agreement against physical fit.
D.4 Testing & Limitations: Over 40 test suites in tests/. Loose winter clothing can skew contour circumferences.